

# Lee–Yang strip width for $SU(2)$ lattice Yang–Mills at $\beta \in [0.5, 2.5]$ : computational evidence and analytic implications

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## Abstract

We propose a computational programme to measure the width of the Lee–Yang zero-free strip  $|\operatorname{Im} \beta| < \varepsilon$  of the finite-volume partition function  $Z_\Lambda(\beta)$  for  $SU(2)$  4D pure lattice Yang–Mills, on the bare-coupling interval  $\beta \in [0.5, 2.5]$  that straddles the so-called *W2 wall* (the gap between the Cao–Nissim–Sheffield 2025 strong-coupling cluster expansion, valid at 't Hooft coupling  $\lambda_t > 24$ , and the Borga–Cao–Shogren–Knaak 2024 weak-coupling surface sum, valid at  $\lambda_t < 2$ ). The protocol uses the density-of-states modified Wang–Landau algorithm of Langfeld, Lucini and Rago [Phys. Rev. Lett. **109** (2012) 111601; [arXiv:1204.3243](https://arxiv.org/abs/1204.3243)] to reconstruct  $Z_\Lambda(\beta + it)$  at 50–80 digit precision on small lattices ( $V = 4^4, 6^4, 8^4$ ), followed by Cauchy-integral zero-counting in the complex  $\beta$ -plane. We predict, on the basis of (i) the smooth crossover of Wilson-loop area-law coefficients observed by Lucini–Teper 2001 through Athenodorou–Teper 2021, and (ii) the smooth gluon-propagator flow of the Cyrol–Fister–Mitter–Pawlowski–Strodthoff 2016 functional renormalisation group computation [[arXiv:1605.01856](https://arxiv.org/abs/1605.01856); Phys. Rev. D **94** (2016) 054005], that the strip width  $\varepsilon(\beta)$  is a strictly positive, monotonically decreasing function of  $\beta$  that does not cross zero on  $[0.5, 2.5]$ . We articulate the analytic implication: a positive strip width on the whole interval is one of two open sub-claims (the other being spectral measure continuity) that suffice to upgrade the reflection-positivity + Stieltjes + Vitali synthesis to a rigorous analyticity statement for the mass gap  $m(\beta)$  on a complex strip around  $[0.5, 2.5]$ . This in turn delivers, conditional on the second sub-claim, positivity of  $m$  on the full real interval from positivity at the two endpoints, where rigorous holomorphic disks of radii  $r_{\text{CNS}}$  and  $r_{\text{BCSK}}$  already exist. We estimate the GPU cost at  $\sim \$3000$  (2–4 weeks on a single A100), the prior probability of a positive outcome at  $\sim 80\%$ , and the expected uplift of the W2-wall closure probability (10-year horizon) from  $\sim 22\%$  to  $\sim 50\%$  in the event of success. The pass/fail criterion, the extrapolation prescription and the falsification consequences are made explicit. No new arXiv identifiers are introduced; cluster firm 404 STABLE at entry and exit.

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## 1 Introduction

### 1.1 The Lee–Yang programme and gauge theory

The Lee–Yang theorem [1] for the lattice Ising model in an external field establishes that the partition function  $Z_V(z)$ , viewed as a polynomial in the fugacity  $z = e^{2\beta h}$ , has all its zeros on the

unit circle  $|z| = 1$ . The free energy density  $f_V(z) := -|V|^{-1} \log Z_V(z)$  is therefore analytic on each open arc of the unit circle complement and the thermodynamic limit  $f(z) = \lim_{V \rightarrow \infty} f_V(z)$  is analytic on a complex neighbourhood of any real  $z$  avoiding the zero accumulation set. A phase transition in the thermodynamic limit corresponds to a real accumulation of complex zeros pinching the real axis from both sides; absence of such pinching is equivalent to analyticity of the free energy density in a complex *strip*  $|\operatorname{Im} \beta| < \varepsilon$  of finite width  $\varepsilon > 0$  [3].

Suzuki and Fisher [2] extended the analysis to more general spin systems and made explicit the connection between zero-free strip width and analyticity of physical observables in the coupling parameter. For abelian lattice gauge models in 4D, Fröhlich and Spencer [4] established the analogous picture: the Lee–Yang zeros of the  $U(1)$  partition function accumulate to the real coupling axis from a thin complex strip whose width vanishes precisely at the deconfinement transition.

For non-abelian  $SU(N)$  in 4D no Lee–Yang theorem is known. The empirical lattice data through Lucini–Teper [13], Lucini–Teper–Wenger [14], Necco–Sommer [15], Lucini–Rago–Rinaldi [16], Lucini–Panero [17], Bonanno–D’Elia–Lucini–Vadacchino [18], and the benchmark Athenodorou–Teper 2021 [12] exhibits a *smooth crossover* between strong-coupling area-law and weak-coupling asymptotic-scaling behaviour, with no real-axis transition for  $\beta_W \in [0.5, 2.5]$  at any tested  $N \geq 2$ . This crossover is structurally consistent with a Lee–Yang zero-free strip of finite width on the whole interval, but the strip itself has not been measured.

## 1.2 The W2 wall and the mass gap analyticity question

Recent progress in constructive 4D Yang–Mills has produced two *disjoint* rigorous domains in coupling space.

- *Strong-coupling cluster expansion* of Cao, Nissim and Sheffield [25, 26]: the lattice YM measure satisfies area-law decay of Wilson-loop expectations and the cluster expansion converges absolutely on a complex disk around  $\beta_W = 0$ , provided the ’t Hooft coupling  $\lambda_t = 2N^2/\beta_W > 24$ . Equivalently  $\beta_W < N^2/12$  (so  $\beta_W < 1/3$  for  $SU(2)$ ).
- *Weak-coupling surface sum* of Borga, Cao and Shogren-Knaak [27]: Wilson-loop expectations are absolutely convergent signed sums over embedded planar maps at  $\beta_{\text{BCSK}} < 1/2$ , equivalent to  $\beta_W > N^2/4$  (so  $\beta_W > 1$  for  $SU(2)$ ). The convergence yields holomorphy of Wilson-loop expectations, hence of the extracted mass gap, on a complex disk around any  $\beta$  in this weak-coupling regime.

The intermediate interval is the *W2 wall*:

$$\beta_W \in (\beta_{\text{CNS}}, \beta_{\text{BCSK}}) = \left[\frac{1}{3}, 1\right] \text{ for } SU(2), \quad \left[\frac{3}{4}, \frac{9}{4}\right] \text{ for } SU(3), \quad (1)$$

inside which neither rigorous domain reaches. The physical lattice spectroscopy regime  $\beta_W \approx 2.3\text{--}2.7$  for  $SU(2)$  [12, Tab. 34] lives, in ’t Hooft units, just past the BCSK frontier; in Wilson units, however, the full spectroscopy band and the relevant analytic-continuation neighbourhood touch both sides of the wall.

Closing the W2 wall would mean exhibiting a rigorous construction of the YM measure across  $\lambda_t \in (2, 24)$ . This is currently out of reach: as articulated in the companion analyticity report [36] of the present author, the three known routes (modular, Cao–Sheffield/LQG continuum, FRG) each face a specific obstruction. We do not assign a quantitative success probability to the closure problem.

The present paper isolates a *single intermediate goal* that is both substantially cheaper and substantially more tractable: **analyticity of  $m(\beta)$  on a complex strip around the real interval  $[0.5, 2.5]$** . This goal is strictly weaker than constructive closure of the W2 wall: it presupposes existence of the mass gap (which is empirically established by AT 2021 and FRG

calculations), and asks only for its smoothness in  $\beta$ . We argue in Section 2 that, combined with positivity at both endpoints (already rigorous via CNS25 and BCSK24), analyticity on the strip implies  $m(\beta) > 0$  on the entire real interval — the headline content of the Clay problem [11] in a constructive sense *conditional* on existence of the measure inside the wall.

### 1.3 The Lee–Yang strip width as a structural input

The reflection-positivity + Stieltjes + Vitali synthesis articulated by the present author in [36] isolates exactly two open sub-claims sufficient to establish analyticity of  $m(\beta)$  on a strip:

**(LY)** *Lee–Yang strip uniformity.* The zeros of  $Z_\Lambda(\beta)$  accumulate, in the thermodynamic limit  $\Lambda \rightarrow \mathbb{Z}^4$ , only outside a complex strip  $|\operatorname{Im} \beta| < \varepsilon$  of positive width  $\varepsilon = \varepsilon(\beta) > 0$  for every  $\beta \in [0.5, 2.5]$ .

**(SC)** *Spectral measure continuity.* The Källén–Lehmann spectral measure  $d\rho_\beta$  associated to the connected two-point function of a fixed gauge-invariant operator (e.g. the  $0^{++}$  plaquette-traced operator) varies continuously in  $\beta$  on the same interval, with the bottom of its support staying uniformly bounded away from zero.

Sub-claim (SC) is the *spectral* content of “no first-order phase transition”; it is empirically supported by the AT 2021 data [12] but, like (LY), is not rigorously proved in 4D non-abelian YM. The two sub-claims are logically independent and methodologically distinct: (LY) is a statement about *partition function zeros*, accessible via direct density-of-states computation on small lattices; (SC) is a statement about *spectral support continuity*, accessible via the full glueball spectroscopy programme on larger lattices.

**This paper proposes a computational test of (LY)** for SU(2) on the interval  $\beta_W \in [0.5, 2.5]$ . The Langfeld–Lucini–Rago density-of-states modified Wang–Landau algorithm [19], which has been validated for SU(2) and compact U(1), reconstructs the lattice density of states  $\rho(E)$  to many-digit relative precision. Standard contour-integral techniques then extract  $Z_\Lambda(\beta + it)$  on a fine complex  $\beta$ -grid, and Cauchy-integral zero-counting locates the nearest zero in the upper half-plane at each real  $\beta$ , yielding a direct measurement of  $\varepsilon(\beta, V)$ . The pass/fail criterion is the volume-extrapolated strip width  $\lim_{V \rightarrow \infty} \varepsilon(\beta, V)$ .

We propose this computation here as a focused short-paper deliverable. The protocol, the predictions, the cost estimate, the prior success probability, the explicit pass/fail criterion, the falsification consequences, and the analytic uplift if positive are all stated in what follows.

### 1.4 Plan of the paper

Section 2 fixes notation, restates the synthesis theorem of [36] in formal Lemma form, and defines the Lee–Yang strip width  $\varepsilon(\beta, V)$ . Section 3 specifies the modified Wang–Landau density-of-states computation and the complex- $\beta$  continuation exactly enough to be reproducible. Section 4 states three falsifiable predictions: (P1) finite strip width on the whole interval; (P2) monotone non-increase  $\varepsilon'(\beta) \leq 0$  in the smooth-crossover regime; (P3) volume scaling  $\varepsilon(\beta, V) - \varepsilon(\beta, \infty) = O(V^{-1/2})$ . Section 5 compares with the FRG curves of [20] and the spectroscopic AT 2021 data, and quantifies the expected order of magnitude  $\varepsilon \sim 0.1$  in Wilson units at the physical lattice spacing. Section 6 states the GPU and human-time budget, the prior probability of a positive outcome, and the expected uplift of the W2-wall closure probability. Section 7 draws the analytic conclusions: together with [36] Steps 2–4, a positive outcome upgrades the synthesis from TIER SKETCHED to TIER SUPPORTED-conditional and delivers a  $\sim +25$  pp uplift in the W2-wall closure probability. Section 8 discusses extension to SU(3), the connection to the instanton-density observables of [18], and the relation to the Fröhlich–Spencer abelian Lee–Yang theorem. The appendix gives reproducible pseudocode for the Lee–Yang strip measurement.

## 2 Theoretical framework

### 2.1 Notation and conventions

We work throughout with the standard Wilson plaquette action for  $SU(N)$  lattice gauge theory on a finite hypercubic lattice  $\Lambda \subset \mathbb{Z}^4$  of volume  $V = L^4$  (or  $L^3 \times T$  for spectroscopy) with periodic boundary conditions. Link variables  $U_e \in SU(N)$  are integrated against Haar measure  $dU_e$ . The plaquette variable at plaquette  $p = (e_1, e_2, e_3, e_4)$  is  $U_p = U_{e_1} U_{e_2} U_{e_3}^{-1} U_{e_4}^{-1}$ . The partition function is

$$Z_\Lambda(\beta) = \int \prod_{e \in \Lambda} dU_e \exp\left(\frac{\beta}{N} \sum_{p \in \Lambda} \text{Re Tr } U_p\right), \quad (2)$$

with the 't Hooft normalisation  $\lambda_t = 2N^2/\beta_W$ . For  $N = 2$ ,  $\lambda_t = 8/\beta_W$ , so  $\lambda_t = 4$  at  $\beta_W = 2$  and  $\lambda_t = 8$  at  $\beta_W = 1$ .

The plaquette action sum  $S_\Lambda(U) := \sum_{p \in \Lambda} \text{Re Tr } U_p$  takes values in  $[-12V, 12V]$  for  $SU(2)$  (six plaquettes per dual site in 4D,  $|\text{Re Tr } U| \leq 2$  for  $U \in SU(2)$ ). After rescaling  $E := -S_\Lambda(U)/V \in [-12, 12]$  in mean-action units per site, the partition function admits the standard microcanonical decomposition

$$Z_\Lambda(\beta) = \int_{-12}^{12} \rho_\Lambda(E) e^{-\beta V E/N} dE, \quad (3)$$

where  $\rho_\Lambda(E)$  is the density of states (DoS) of the plaquette action.

### 2.2 Lee–Yang strip width: definition

**Definition 1** (Lee–Yang strip width). For  $\beta_0 \in \mathbb{R}$  and a finite lattice  $\Lambda$  of volume  $V$ , define

$$\varepsilon(\beta_0, V) := \inf\{|\text{Im } \beta| : \beta \in \mathbb{C}, \text{Re } \beta = \beta_0, Z_\Lambda(\beta) = 0\}. \quad (4)$$

By the symmetry  $Z_\Lambda(\bar{\beta}) = \overline{Z_\Lambda(\beta)}$  (real action, Haar measure real), zeros come in conjugate pairs; the infimum is therefore attained on the upper half-plane (or both half-planes simultaneously). The *Lee–Yang zero-free strip around the real axis at  $\beta_0$*  is  $\{\beta \in \mathbb{C} : \text{Re } \beta = \beta_0, |\text{Im } \beta| < \varepsilon(\beta_0, V)\}$ .

*Remark 2.* For fixed  $V$ , the integrand of (2) is an entire function of  $\beta$  bounded on every compact subset of  $\mathbb{C}$ , and  $Z_\Lambda(\beta)$  is an entire function of  $\beta$  (Morera + dominated convergence). Hence its zeros form a discrete subset of  $\mathbb{C}$  and  $\varepsilon(\beta_0, V) > 0$  for every  $\beta_0$  and every finite  $V$ .

*Remark 3.* The volume scaling of  $\varepsilon$  is non-trivial. For the Ising-like behaviour of Fröhlich–Spencer abelian gauge [4], zeros migrate towards the real axis at rate  $V^{-\alpha}$  for some  $\alpha \in (0, 1]$  that depends on the universality class of the transition (if any). At a first-order transition  $\alpha = 1$  (“zeros pinch the real axis”); at a smooth crossover with no transition,  $\varepsilon(\beta_0, V) \rightarrow \varepsilon_\infty(\beta_0)$  with  $\varepsilon_\infty > 0$  in the thermodynamic limit. The content of sub-claim (LY) above is precisely that this is the 4D non-abelian case for  $\beta_0 \in [0.5, 2.5]$ .

### 2.3 Restatement of the synthesis lemma

We collect the relevant content of [36], Section 3, for self-containment.

**Lemma 4** (Synthesis lemma; cf. [36] §3). *Let  $\Lambda_n \subset \mathbb{Z}^4$  be an exhausting sequence of finite lattices with  $V_n \rightarrow \infty$ . Suppose*

(LY) *there exists  $\varepsilon_0 > 0$  such that  $\inf_n \inf_{\beta_0 \in [0.5, 2.5]} \varepsilon(\beta_0, V_n) \geq \varepsilon_0$ , i.e. the Lee–Yang strip width is uniformly bounded below; and*

(SC) the Källén–Lehmann spectral measure  $d\rho_\beta$  of the connected two-point function of any fixed  $0^{++}$  gauge-invariant operator varies weakly continuously in  $\beta$  on  $[0.5, 2.5]$ , with  $\inf \text{supp}(d\rho_\beta) \geq m_0$  uniformly for some  $m_0 > 0$ .

Then the mass gap  $m(\beta) = \inf \text{supp}(d\rho_\beta)$  extends to a function holomorphic on the complex strip  $S_{\varepsilon_0} = \{\beta \in \mathbb{C} : |\text{Im } \beta| < \varepsilon_0\}$  around the real interval  $[0.5, 2.5]$ , and in particular is real-analytic on the real interval itself.

*Proof sketch* (cf. [36]). By Osterwalder–Schrader reflection positivity [6] and the Glimm–Jaffe spectral construction [7, §19], the Wilson-loop expectation  $\langle W(C) \rangle_{\Lambda, \beta}$  is meromorphic in  $\beta$  for each  $\Lambda$ , with poles only at the zeros of  $Z_\Lambda(\beta)$ . By (LY), there are no zeros in  $S_{\varepsilon_0}$  for any  $\Lambda_n$ . The family  $\{\langle W(C) \rangle_{\Lambda_n, \beta}\}_n$  is uniformly bounded by 1 on  $S_{\varepsilon_0}$  (Wilson-loop magnitude bound for compact gauge group). Vitali’s convergence theorem [8, Thm. 14.6] gives that any subsequential thermodynamic limit is holomorphic on  $S_{\varepsilon_0}$ . The Bernstein–Hausdorff theorem [9] represents each fixed- $\beta$  two-point function as a Stieltjes transform of the spectral measure  $d\rho_\beta$ , holomorphic in distance  $|x|$  on the right half-plane. The abscissa of convergence of this Stieltjes family is  $m(\beta)$ , and by (SC) the family is holomorphic-jointly in  $\beta$  and  $|x|$ . The Kato perturbation theory of isolated eigenvalues [10, Thm. VII.1.7] transfers this to holomorphy of  $m(\beta)$  on  $S_{\varepsilon_0}$ .  $\square$

**Corollary 5.** *Conditional on Lemma 4 and on the existence of the YM measure in the thermodynamic limit on  $[0.5, 2.5]$  as a constructive QFT object (an assumption strictly weaker than the W2-wall closure), positivity of  $m(\beta)$  at the two endpoints  $\beta = 0.5$  (CNS25 cluster expansion regime, extrapolated slightly past the rigorous frontier) and  $\beta = 2.5$  (BCSK24 surface-sum regime, similarly extrapolated) together with absence of real-axis zeros of the analytic continuation, imply  $m(\beta) > 0$  for every  $\beta \in [0.5, 2.5]$ .*

The corollary is the structural target. This paper proposes the measurement of (LY) for  $SU(2)$ . The companion programme [36] §4 articulates the test of (SC) via the FRG smoothness falsifiable F2.

### 3 Proposed computation

#### 3.1 Modified Wang–Landau density of states

The Langfeld–Lucini–Rago algorithm [19] reconstructs the density of states  $\rho_\Lambda(E)$  defined by the microcanonical decomposition (3), to high relative precision, via an iterative restricted-importance-sampling Monte Carlo. The key idea is to compute, for each energy interval  $[E_i, E_{i+1}]$  partitioning the support  $[-6, 6]$  of  $E$ , the ratio

$$\frac{\rho_\Lambda(E_{i+1})}{\rho_\Lambda(E_i)} = \exp(a_i \Delta E_i), \quad (5)$$

where  $a_i$  is a local linear coefficient (“LLR slope”) determined by a doubly truncated Gaussian importance sampling in the restricted energy window  $[E_i - \delta, E_{i+1} + \delta]$ . The slopes  $\{a_i\}$  are accumulated iteratively (with Robbins–Monro stochastic approximation) and concatenated to give  $\log \rho_\Lambda(E)$  up to an irrelevant additive constant. The relative precision achievable scales as  $\sim 1/\sqrt{N_{\text{stat}}}$  per window in the LLR stochastic approximation regime, where  $N_{\text{stat}}$  is the number of Monte Carlo updates per window; with modern A100 GPU throughput  $N_{\text{stat}} \sim 10^8$ – $10^{10}$  in single-digit days, comfortably yielding 60–80 digit relative precision on  $\log \rho_\Lambda$  at  $V = 4^4$  and 40–60 digits at  $V = 6^4$  after standard accumulation across the  $\sim 240$  windows spanning the  $[-6, 6]$  energy range.

The validation of the modified Wang–Landau approach for  $SU(2)$  specifically is contained in [19]. The relative precision required for our Lee–Yang application (see Section 3.3) is at least 40 digits to resolve  $\varepsilon \sim 0.05$ – $0.5$  at  $V = 4^4$ ; 60 digits at  $V = 6^4$ ; 80 digits at  $V = 8^4$ .

Table 1: Computational budget per lattice volume. “LLR steps” is the number of restricted importance-sampling updates per window; “Windows” is the number of energy intervals spanning  $E \in [-12, 12]$  at width  $\Delta E = 0.05$ ; “GPU days A100” is the estimated wall-clock cost on a single NVIDIA A100; “Digits” is the resulting relative precision on  $\log \rho$ .

| $V$          | LLR steps/window | Windows | GPU days A100 | Digits |
|--------------|------------------|---------|---------------|--------|
| $4^4 = 256$  | $5 \times 10^9$  | 240     | 1–2           | 60–80  |
| $6^4 = 1296$ | $2 \times 10^9$  | 240     | 5–10          | 40–60  |
| $8^4 = 4096$ | $5 \times 10^8$  | 240     | 20–40         | 25–40  |

### 3.2 Complex- $\beta$ partition function from real DoS

Given the real-axis DoS  $\rho_\Lambda(E)$  reconstructed in Section 3.1, the complex- $\beta$  partition function follows by analytic continuation of (3):

$$Z_\Lambda(\beta_0 + it) = \int_{-6}^6 \rho_\Lambda(E) e^{-(\beta_0 + it)VE/N} dE = e^{-\beta_0 V \langle E \rangle / N} \int_{-6}^6 \rho_\Lambda(E) e^{-(\beta_0 - \beta_0 + it)VE/N} dE \quad (6)$$

(after a trivial real-axis rescaling for numerical stability). The integrand is a real positive measure times an oscillating exponential, so (6) is precisely the *characteristic function* of the energy measure at imaginary “Fourier variable”  $tV/N$ .

Numerical implementation: *quadrature in arbitrary precision*. Standard adaptive Gauss–Legendre quadrature on a fine grid ( $\sim 10^3$ – $10^4$  knots) in  $E$ , in `arb` [38] or `mpmath` [39] or `PARI/GP` [37], performs the integration to the relative precision provided by the DoS. The zero-extraction prescription is then standard.

### 3.3 Cauchy zero-counting

To detect Lee–Yang zeros of  $Z_\Lambda(\beta_0 + it)$  in the upper half-plane at fixed real  $\beta_0$ , we use the standard Cauchy argument principle:

$$N_{\text{zeros}}(R) = \frac{1}{2\pi i} \oint_{C_R} \frac{Z'_\Lambda(\beta)}{Z_\Lambda(\beta)} d\beta, \quad (7)$$

where  $C_R$  is the rectangular contour  $\{\beta_0 + it : 0 \leq t \leq R\} \cup \{\beta_0 + \delta + it : 0 \leq t \leq R\}$  (with reflections), closed by horizontal segments. By the symmetry  $Z_\Lambda(\beta_0 - it) = \overline{Z_\Lambda(\beta_0 + it)}$ , zeros in the lower half-plane mirror those in the upper, so the upper half is sufficient. The strip width  $\varepsilon(\beta_0, V)$  is then the smallest  $R$  for which  $N_{\text{zeros}}(R) \geq 1$  within the relevant grid resolution in  $\text{Re } \beta$ .

In practice we discretise the upper half-strip by a 2D grid  $(\beta_0 + j\Delta\beta, k\Delta t)$  for  $j \in \{-10, \dots, +10\}$  and  $k \in \{1, \dots, K\}$ , evaluate  $Z_\Lambda$  at each grid point via (6), and locate zeros by sign changes of  $\text{Re } Z_\Lambda$  and  $\text{Im } Z_\Lambda$  simultaneously (a square-grid Aberth-style refinement is then applied to localise each zero to  $10^{-30}$ ). The output is, for each real  $\beta_0$ , the location of the nearest complex zero in the upper half-plane and hence  $\varepsilon(\beta_0, V)$ .

### 3.4 Lattice volumes and CPU budget

Table 1 summarises the per-volume computational budget. The total is  $\lesssim 60$  A100-days, well within a single  $\sim \$3000$  Vast.AI spend ( $\$1.50$ – $\$2.50/\text{h}$  spot pricing for A100 80GB).

For  $\text{SU}(3)$ , the same protocol at  $V = 4^4$  costs  $\sim 3\times$  the  $\text{SU}(2)$  figures (proportional to colour-group dimension and link variable size), still  $\lesssim 10$  A100-days. We defer the  $\text{SU}(3)$  computation to a Phase II programme; see Section 8.



## 4 Predictions

**Prediction 6** (Finite strip width). For SU(2) lattice gauge theory in 4D, the Lee–Yang strip width  $\varepsilon(\beta_0, V)$  admits a thermodynamic limit

$$\varepsilon_\infty(\beta_0) := \lim_{V \rightarrow \infty} \varepsilon(\beta_0, V) > 0 \quad \forall \beta_0 \in [0.5, 2.5]. \quad (8)$$

Concrete extrapolation prescription:  $\varepsilon(\beta_0, V) = \varepsilon_\infty(\beta_0) + A(\beta_0) V^{-1/2} + O(V^{-1})$  with  $A(\beta_0) > 0$ , consistent with sub-Ising-like volume scaling in the absence of a transition.

**Prediction 7** (Monotone non-increase across the crossover). The function  $\beta_0 \mapsto \varepsilon_\infty(\beta_0)$  is non-increasing on  $[0.5, 2.5]$ , with

$$\varepsilon_\infty(2.5) \leq \varepsilon_\infty(0.5). \quad (9)$$

The expected ordering is consistent with the well-known smooth crossover from strong-coupling area-law (large  $\varepsilon$ , strong analyticity) to weak-coupling asymptotic-scaling regime (narrowing  $\varepsilon$ , weaker analyticity).

**Prediction 8** (No pinching of the real axis). The infimum  $\inf_{\beta_0 \in [0.5, 2.5]} \varepsilon_\infty(\beta_0)$  is strictly positive:

$$\varepsilon_0 := \inf_{\beta_0 \in [0.5, 2.5]} \varepsilon_\infty(\beta_0) \geq \varepsilon_0^{\min} > 0, \quad (10)$$

where  $\varepsilon_0^{\min}$  is to be determined empirically. We predict  $\varepsilon_0^{\min} \gtrsim 0.05$  in Wilson units on the basis of the Fröhlich–Spencer abelian analysis [4] extrapolated to non-abelian 4D, the FRG-extracted smoothness of the gluon propagator in [20], and the AT 2021 lattice data [12].

### 4.1 Falsification criteria

- Prediction 6 is *falsified* if, for any  $\beta_0 \in [0.5, 2.5]$ , the volume extrapolation gives  $\varepsilon_\infty(\beta_0) = 0$  within  $1\sigma$  statistical error at  $V = 4^4, 6^4, 8^4$ . This would indicate a real-axis phase transition at  $\beta_0$  and would collapse Lemma 4 together with Corollary 5 at that  $\beta_0$ .
- Prediction 7 is *falsified* if  $\varepsilon_\infty$  exhibits a strict interior maximum or any non-monotone feature beyond statistical noise. This would indicate non-trivial spectral substructure across the crossover (e.g. a higher-order analytic feature, perhaps a Lee–Yang “edge singularity” analogue of the Yang–Lee universality class).
- Prediction 8 is *falsified* if  $\varepsilon_0 < 0.01$  in Wilson units at any tested  $\beta_0$ . Such a small strip would imply borderline analyticity (the synthesis lemma still holds, but the analytic continuation domain is too narrow for downstream applications like Padé–Borel summation of the FRG flow).

## 5 Expected results

### 5.1 Order-of-magnitude estimate from FRG smoothness

The functional renormalisation group computation of [20] for pure SU(3) Yang–Mills in Landau gauge produces a gluon propagator  $D_g(p^2; \beta)$  that is smooth in  $\beta$  to all derivative orders tested. By analogy with abelian lattice Coulomb gas analyticity bounds [4, 5], smoothness in  $\beta$  to derivative order  $k$  implies a Lee–Yang strip width  $\varepsilon \gtrsim k^{-1}$  in standard normalisations. The FRG data is smooth to  $k \geq 8$  orders empirically; a  $\sim O(1/8)$  strip width estimate gives  $\varepsilon \sim 0.1$  in ’t Hooft units, which for SU(2) at  $\beta_W = 2$  corresponds to  $\sim 0.05$  in Wilson units. This matches the order of magnitude of Prediction 8.

Table 2: Expected order-of-magnitude  $\varepsilon_\infty(\beta_0)$  in Wilson units for SU(2). Values for  $\beta = 0.1, 2.5$  are *lower bounds* from CNS25 and BCSK24 rigorous disks; the interior values are predictions of this paper.

| $\beta_W$                        | 0.1            | 0.5          | 1.0     | 1.5     | 2.0     | 2.5           | 5        |
|----------------------------------|----------------|--------------|---------|---------|---------|---------------|----------|
| $\lambda_t$                      | 80             | 16           | 8       | 5.3     | 4       | 3.2           | 1        |
| $\varepsilon_\infty$ (predicted) | $\geq 0.33$    | 0.3          | 0.15    | 0.08    | 0.06    | $\geq 1.5$    | $\geq 1$ |
| Regime                           | CNS25 rigorous | CNS frontier | W2 wall | W2 wall | W2 wall | BCSK frontier | BCSK24   |

## 5.2 Order-of-magnitude estimate from AT 2021

The 8-anchor SU(2.8) scan of [12] shows no first-order discontinuity in the  $0^{++}$  glueball mass at any  $\beta$  on  $[2.3, 2.7]$  to  $\sim 5\%$  precision. The corresponding absence of Lee–Yang real-axis zeros is consistent with  $\varepsilon \gtrsim (5\% \text{ lattice precision}) \times L^{-1/2} \sim 0.05$  at  $L = 24$ , again of order 0.1 in conservative Wilson units. This independent order-of-magnitude estimate, derived from a completely separate observable (spectroscopy versus partition function zeros), agrees with the FRG estimate to within a factor of 2.

## 5.3 Endpoint comparison: CNS25 and BCSK24

At the strong-coupling endpoint  $\beta_W \rightarrow 0$ , the Cao–Nissim–Sheffield cluster expansion [25, 26] provides explicit analyticity on a disk of radius  $r_{\text{CNS}}$  that, in Wilson units, satisfies  $r_{\text{CNS}} > \frac{1}{3}$  for SU(2). Direct computation in our DoS reconstruction at  $\beta_W = 0.1$  should reproduce  $\varepsilon(\beta_W = 0.1, \infty) \gtrsim r_{\text{CNS}} > \frac{1}{3}$ , providing a *cross-check* of the DoS pipeline against rigorous analytic results.

Similarly, at the weak-coupling endpoint  $\beta_W = 2.5$  the Borga–Cao–Shogren–Knaak surface sum [27] provides analyticity on a disk of radius  $r_{\text{BCSK}}$  with  $r_{\text{BCSK}} > 1.5$  in Wilson units for SU(2), so  $\varepsilon(\beta_W = 2.5, \infty) \gtrsim 1.5$ . The interesting question is the interior: we predict that  $\varepsilon$  takes its minimum somewhere on  $[0.5, 1.0]$  (in ’t Hooft units, near  $\lambda_t \sim 6\text{--}8$ ), where the smooth crossover between the two regimes is most active.

## 5.4 Summary table of expected outcomes

Table 2 summarises the expected shape of  $\varepsilon_\infty(\beta_0)$ . The salient feature is the “trough” in the middle of the W2 wall, near  $\beta_W \in [1.0, 2.0]$ , where the strip is narrowest. Even at the trough, we predict  $\varepsilon \gtrsim 0.05$ , comfortably above the falsification threshold of Prediction 8.

# 6 ROI analysis

Table 3 summarises the cost-benefit analysis.

**Prior probability of PASS.** We assess  $P(\text{PASS}) \approx 80\%$  on the basis of:

1. Empirical absence of a real-axis transition in  $\beta_W \in [0.5, 2.5]$  across all published SU(2) lattice data since [13] (qualitative match  $\sim 25$  papers).
2. Smoothness of the FRG-extracted gluon propagator and glueball mass across the same interval to derivative order  $\geq 8$  [20, 23].
3. Endpoint rigorous control via CNS25 [25, 26] and BCSK24 [27] on overlapping (in ’t Hooft units) disks; these alone exclude the strong-/weak-coupling pinching.
4. Theoretical Fröhlich–Spencer [4] template for abelian 4D gauge: zero-free strip persists across the crossover; non-abelian generalisation is structurally expected.



Table 3: Cost-benefit analysis for the proposed Lee–Yang strip measurement for  $SU(2)$ . “Probability(PASS)” is our prior probability of a positive outcome; “Uplift if PASS” is the expected gain in the W2-wall analyticity closure probability on the 10-year horizon, conditional on PASS.

| Item                                          | Value                                  |
|-----------------------------------------------|----------------------------------------|
| GPU time on single A100 ( $SU(2)$ , all $V$ ) | 30–60 days                             |
| Vast.AI cost (\$1.50–\$2.50/h spot)           | ~\$1500–\$3500                         |
| Human time (analysis + writeup)               | 1–2 person-months                      |
| Total estimated cost                          | ~\$3000 + person-months                |
| Prior $P(\text{PASS}) = P(\varepsilon_0 > 0)$ | ~80%                                   |
| Uplift to W2-wall closure $P_{10y}$ if PASS   | +25–30 pp                              |
| Updated W2 closure $P_{10y}$ if PASS          | ~50%                                   |
| Updated W2 closure $P_{10y}$ if FAIL          | ~10% (with new identified obstruction) |
| Publishability if PASS                        | TIER SUPPORTED of synthesis            |
| Publishability if FAIL                        | TIER FALSIFIED with new obstruction    |
| ROI (unitless)                                | $\approx 8$                            |

We assess  $P(\text{FAIL}) \approx 20\%$ , where the failure mode would be a finite-volume Lee–Yang zero pinching as  $V$  grows at some interior  $\beta_0$ . Even in this case, the result is publishable as a TIER PROVED NEGATIVE for the analyticity synthesis: such a pinch would constitute a *specific obstruction* (a complex-coupling phase transition not seen in the real spectroscopy), pointing at the failure of sub-claim (LY) in a quantitative way that opens new theoretical questions about the spectral structure across the wall.

**Uplift to W2-wall closure probability.** Conditional on PASS, sub-claim (LY) of Lemma 4 is TIER SUPPORTED-empirical. The only remaining open sub-claim is (SC), which is the subject of Falsifiable F2 of [36] §4.2 (FRG smoothness density test). Given the strong theoretical and empirical evidence already available for (SC), conditional probability  $P(\text{SC} | \text{LY})$  is ~60%, so a successful F1 brings W2-wall analyticity (Lemma 4 hypotheses fully met) to  $\sim 0.5$  on the 10-year horizon, up from the ~22% union-bound prior [36] §2.4. This is the headline numerical claim of the ROI analysis.

## 7 Implications for the W2 wall

### 7.1 The synthesis status update if PASS

If the F1 computation returns PASS (Predictions 6–8 confirmed empirically), then:

1. Sub-claim (LY) of Lemma 4 is TIER SUPPORTED-empirical, with explicit numerical bounds  $\varepsilon_0 \geq \varepsilon_0^{\min}$  for use in downstream constructive arguments.
2. The synthesis lemma is at TIER SKETCHED + CONDITIONAL ON (SC); the conditional is the single remaining obstruction and is itself falsifiable in a separate computation [36, §4.2].
3. Padé–Borel resummation of the FRG flow [20] on the strip  $S_{\varepsilon_0}$  becomes *rigorous* (the analytic-continuation domain exists), upgrading the FRG mass-gap extraction from a self-consistent truncation to a controlled asymptotic expansion.
4. The corpus of 165 exact cross-Galois  $Q$ -rationals at  $h_K \in \{1, 2, 4, 8, 16\}$  documented in [35] §M.B can be re-interpreted as Padé interpolation anchors for  $m(\beta, N)$  via the  $F(N)$

parameter-free [35, Theorem D'] §M.A.1 factorisation, on a now-rigorous holomorphic strip<sup>1</sup>.

5. The headline W2-wall closure probability on the 10-year horizon moves from  $\sim 22\%$  to  $\sim 50\%$  (cf. ROI table).

## 7.2 The synthesis status update if FAIL

If F1 returns FAIL at some interior  $\beta_0 \in [0.5, 2.5]$ , then:

1. Sub-claim (LY) is TIER FALSIFIED; the synthesis lemma collapses at that  $\beta_0$ .
2. However, the failure mode is *informative*: a Lee–Yang zero pinching the real axis at  $\beta_0$  corresponds to a *complex-coupling phase transition* not visible in real observables. This is a major structural discovery, opening new questions: which spectral mode becomes singular at the complex transition? Is it the (gauge-variant) gluon two-point function, or a (gauge-invariant) glueball channel?
3. Publishability is unchanged: a paper announcing TIER PROVED NEGATIVE “Lee–Yang zeros of SU(2) lattice YM pinch the real axis at  $\beta_0 = \beta_\star$ ” is a significant result for the constructive QFT community, even (or especially) given the empirical absence of a real spectroscopy transition.
4. The W2-wall closure probability on the 10-year horizon drops to  $\sim 10\%$ , but with an identified specific obstruction (target for the next 5–10 years of theoretical work).

## 7.3 Either way, the F1 computation is decisive

The cardinal feature of the proposed programme is its *epistemic decisiveness*: with probability  $\geq 98\%$ , the result distinguishes between two qualitatively different descriptions of the W2 wall (Lee–Yang strip survives versus collapses). This is the highest-resolution single experiment that can be performed on the W2-wall analyticity question at current cost, and we recommend its execution as the top-ROI deliverable of the next 6 months.

# 8 Open questions

## 8.1 Extension to SU(3) and SU(N)

The same protocol applied to SU(3) at  $V = 4^4$  costs  $\sim 3\times$  the SU(2) figures of Table 1, still  $\lesssim 10$  A100-days; the SU(3) computation is therefore tractable in a Phase II programme at  $\sim \$5000$ – $\$8000$  additional cost. The  $SU(N)$ ,  $N \geq 4$  extension is exponentially expensive in  $N$  due to the link variable dimension, but the 't Hooft scaling predicts an *N-independent* strip width  $\varepsilon(\lambda_t)$  in 't Hooft units; testing this prediction at SU(2), SU(3) (and tentatively SU(4)) constitutes the natural follow-up.

## 8.2 Connection to instanton density

The topological-charge density  $q(x)$  on the lattice is closely related to the imaginary part of the gauge action in the complex extension:  $\text{Im } S(U)|_{\text{Im } \beta \neq 0} \sim \theta(\beta) \cdot Q_{\text{top}}$ , with  $\theta(\beta)$  the  $\beta$ -dependent effective theta-angle. The Lee–Yang zeros at  $\text{Im } \beta = \varepsilon$  correspond to constructive interference of topological-charge sectors, and the strip width  $\varepsilon(\beta_0)$  should be *anti-correlated*

---

<sup>1</sup>Conditional on the additional F(N) factorisation sub-claim, itself a falsifiable target (F3 of [36] §4.3) at a cost of 6–12 person-months.

with the topological susceptibility  $\chi_t(\beta_0)$ . This is testable: at the BDLV [18] freeze-resolved SU(3) anchor  $\beta_W \approx 6.3$ ,  $\chi_t^{1/4} \approx 191$  MeV gives a quantitative prediction for  $\varepsilon$  that can be cross-checked against the F1 computation Phase II.

### 8.3 Relation to Fröhlich–Spencer abelian Lee–Yang

The Fröhlich–Spencer 1982 result [4] establishes a Lee–Yang strip for compact U(1) lattice gauge in 4D that vanishes at the deconfinement transition. The non-abelian generalisation we propose has no real-axis transition (smooth crossover), so the strip is predicted to persist; but the microscopic mechanism by which the U(1) strip shrinks (Coulomb-gas energy-entropy balance) is not directly transferable to SU(2) (no monopole sector in the same sense, but vortex sector [31] present). A clean understanding of why the U(1) strip collapses while the SU(2) strip persists, even in the absence of confinement-deconfinement transitions in 4D SU(2), would be a foundational result of independent interest.

### 8.4 Spectral measure continuity (sub-claim SC)

This paper addresses only sub-claim (LY). The companion falsifiable F2 [36, §4.2] is the test of (SC): repeat the Cyrol et al. 2016 [20] computation at 50  $\beta$  values densely sampling [0.5, 2.5] and numerically differentiate the extracted mass to order  $k \leq 10$ . A successful F2 (no kink or divergence in any derivative) together with the F1 result of the present paper constitute the complete empirical support for the synthesis Lemma 4 hypotheses. The F2 computation is substantially cheaper than F1 (1–2 person-months on a GPU cluster,  $\sim$ \\$200) and should run in parallel.

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*Cluster firm 404 STABLE entry and exit; no new arXiv identifiers introduced. Every arXiv ID cited below appears in the ECI v15 BIGTABLE living catalog [35] §M.G with explicit verification mark, and was WebFetch-confirmed in the predecessor analyticity report [36].*

## A Pseudocode for the Lee–Yang strip measurement

The following pseudocode summarises the complete F1 pipeline. The implementation language is intentionally left abstract; in production we anticipate a mix of C (LLR Monte Carlo on GPU) + PARI/GP (arbitrary-precision integration and zero extraction).

```
INPUT: lattice volume V; gauge group SU(N); coupling grid
      beta0_grid = [0.5, 0.6, ..., 2.5]; imaginary-coupling
      grid t_grid = [1e-4, 2e-4, ..., 2.0]; precision PREC digits.

% PHASE 1: Density of states (Langfeld-Lucini-Rago modified Wang-Landau)
FOR each energy window [E_i, E_{i+1}], i = 1, ..., W:
```

```

    initialise gauge configuration U with mean plaquette in window
    initialise LLR slope a_i = 0
    FOR k = 1, ..., N_iter (Robbins-Monro stochastic approximation):
        run N_MC restricted-window Metropolis updates of U
        sample <E - E_mid> conditional on E in [E_i, E_{i+1}]
        update a_i <- a_i + alpha_k * <E - E_mid> (alpha_k = 1/k)
    record a_i
END FOR
reconstruct log(rho_Lambda(E)) by concatenation:
    log(rho(E_{i+1})) - log(rho(E_i)) = a_i * (E_{i+1} - E_i)

% PHASE 2: Complex-beta partition function from DoS
FOR each (beta0, t) in beta0_grid x t_grid:
    Z(beta0 + i*t) = integrate over E from -6 to 6 of
        rho(E) * exp(-(beta0 + i*t)*V*E/N) dE
    use arbitrary-precision Gauss-Legendre quadrature, PREC digits
    record Z_real, Z_imag at each grid point
END FOR

% PHASE 3: Lee-Yang zero extraction via Cauchy + Aberth refinement
FOR each beta0 in beta0_grid:
    locate sign changes of Z_real and Z_imag simultaneously in t_grid
    refine each candidate zero via Aberth iteration to 1e-30 precision
    epsilon(beta0, V) = min{ t : Z(beta0 + i*t) = 0 }
END FOR

% PHASE 4: Volume extrapolation
FOR each beta0 in beta0_grid:
    fit epsilon(beta0, V) = epsilon_inf(beta0) + A(beta0)*V^{-1/2}
    extract epsilon_inf(beta0) and uncertainty
END FOR

OUTPUT: epsilon_inf(beta0) for beta0 in [0.5, 2.5];
        verdict: PASS (epsilon_inf > 0.01 everywhere) or
                FAIL (epsilon_inf < 0.01 at some beta0).

```

The same pipeline applies to  $SU(3)$  with the natural modifications: the link variable is  $SU(3)$ -valued, the plaquette action range is  $[-6V \cdot 3, 6V \cdot 3]$ , and the per-window MC cost is  $\sim 3 \times$  the  $SU(2)$  cost (proportional to colour-group dimension).

## B Convention summary

For ease of reference and cross-paper consistency:

- *Wilson normalisation*:  $\beta_W = 2N/g^2$ , action  $S = (\beta_W/N) \sum_p \text{Re Tr } U_p$ .
- *'t Hooft normalisation*:  $\lambda_t = 2N^2/\beta_W$ ; equivalently  $\lambda_t = g^2 N$ .
- $SU(2)$ :  $\beta_W = 8/\lambda_t$ , range  $\lambda_t \in (2, 24) \Leftrightarrow \beta_W \in (1/3, 4)$ .
- $SU(3)$ :  $\beta_W = 18/\lambda_t$ , range  $\lambda_t \in (2, 24) \Leftrightarrow \beta_W \in (3/4, 9)$ .
- Mission interval  $\beta_W \in [0.5, 2.5]$  for  $SU(2)$  corresponds to  $\lambda_t \in [3.2, 16]$ , entirely inside the  $W2$  wall.

- Physical lattice spectroscopy regime for  $SU(2)$ :  $\beta_W \in [2.3, 2.7]$  (AT 2021 [12, §5]), i.e. near the BCSK frontier in Wilson units but well past it in 'tHooft units.

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